

Note: This will be a closed-book exam. You may use one two-sided, letter-sized sheet of notes and a scientific calculator. Smartphones, tablets, smartwatches, laptops, and notebooks will not be allowed. However in the mock exam you are free to use any resources you like.

1. Consider the following simplified version of the broken telephone game. A first player secretly communicates a message $m_1 \in \{0, 1, 2\}$ to a second player with probabilities

$$P(m_1 = 0) = P(m_1 = 1) = P(m_1 = 2) = \frac{1}{3}$$

The second player then secretly communicates a message $m_2 = m_1 + n$ to a third (final) player

$$P(n = 0) = \frac{2}{3} \quad P(n = +1) = \frac{1}{3}$$

- a. What is the optimal Bayes classification of the message received by the last player?

Answer:

- b. What is the expected error probability?

Answer:

2. Many air travelers have waited expectantly in an airport baggage claim area, watching for their bags to drop down the chute into the circulating luggage carousel (figure below). This situation presents opportunities for probabilistic reasoning as well as perceptual inference. Let us suppose that you are engaged in this ritual of modern-day air travel along with 99 other passengers from your flight, each of whom, like you, checked one item of luggage. A recording piped through the speakers reminds you that *“Many bags look alike. Please check your bag carefully before exiting the terminal.”* Indeed, your bag is one of the most popular models on the market, a medium-sized black rectangular case used by 5 percent of all travelers. Given your distance from the luggage chute, you can discern only the shape, size and color of the bags that emerge, not any detailed markings on the bags. Now let us suppose that the first bag from your flight to enter the luggage carousel indeed has the same shape, size, and color as your bag.



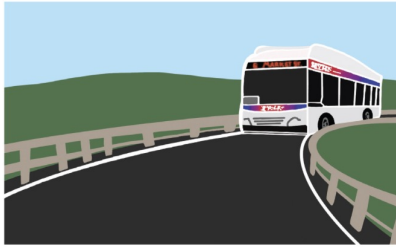
- a. Is it yours? Explain.

Answer:

- b. How the probability changes as more look-alikes appear?

Answer:

3. Suppose you are waiting to catch a particular bus in a city that has just ten bus routes; the route followed by each bus is indicated by an integer in the corner of its front display. You see the bus below from a distance, and naturally wonder whether this is the bus you are waiting for.



- a. Based on the visual image of the difficult-to-discern bus route number, and your intuitive understanding of how different route numbers might appear, construct a plausible likelihood function that plots $p(\text{visual observation} \mid \text{hypothesized bus route})$, for all numbers from 1 to 10. (Assume that the written destination does not mean anything to you.)

Answer:

- b. As it turns out, you happen to know that only buses 3, 4, 5, and 6 travel down the street you are on. Furthermore, you know that buses 3 and 4 come twice as frequently as buses 5 and 6. Based on this background knowledge, construct your prior distribution for the bus number.

Answer:

- c. Use Bayes' rule to calculate your posterior distribution for the number of the bus.

Answer: